

Seminar 7

Scalar maps

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. We define the iterates of f as

$$f^2 = f \circ f, \quad f^3 = f \circ f^2, \quad \dots, \quad f^k = f \circ f^{k-1}, \quad \dots \quad \forall k \geq 2$$

with the convention that f^0 is the identity and $f^1 = f$.

We say that η^* is a fixed point of f when $f(\eta^*) = \eta^*$.

Note that a fixed point of f is a fixed point of f^k , $\forall k \geq 2$.

We say that η^* is an attractor for f when for each η sufficiently close to η^* we have $\lim_{k \rightarrow \infty} f^k(\eta) = \eta^*$.

For an attractor η^* the set $A_{\eta^*} = \{ \eta \in \mathbb{R} : \lim_{k \rightarrow \infty} f^k(\eta) = \eta^* \}$ is called the basin of attraction of η^* .

Theorem 1 Assume that $f \in C^1(\mathbb{R})$ and that η^* is a fixed point.

If $|f'(\eta^*)| < 1$ then η^* is an attractor (asymptotically stable).

If $|f'(\eta^*)| > 1$ then η^* is unstable (repellor).

We say that η^* is a p-periodic point of f when $f^p(\eta^*) = \eta^*$ and $f^k(\eta^*) \neq \eta^*$ for any $k \in \{1, 2, \dots, p-1\}$.

We say that a p-periodic point η^* of f is attractor/repellor ~~if~~ when η^* is a fixed point attractor/repellor for f^p .

The (positive) orbit of an initial state $\eta \in \mathbb{R}$ is

$$\mathcal{O}_\eta^+ = \{ \eta, f(\eta), \dots, f^k(\eta), \dots \}.$$

The unique solution of the IVP $x_{k+1} = f(x_k)$, $x_0 = \eta$

denoted $\varphi(k, \eta)$ can be written also as $\varphi(k, \eta) = f^k(\eta)$
 $\forall k \geq 0, \forall \eta \in \mathbb{R}$.

Exercise 1. We consider the map $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x - \frac{1}{4}(x^2 - 2)$.

a) Find the fixed points of f and study their stability.

b) Write the consequences of a) for the IVP

$$x_{k+1} = x_k - \frac{1}{4}(x_k^2 - 2), \quad x_0 = \eta.$$

c) Using the stair-step diagram study the behaviour of the sequence $(x_k)_{k \geq 0}$ given at b) for any initial value $\eta \in \mathbb{R}$.

Solution. a) The fixed points are the solutions of the equation $f(x) = x$. So we have to solve

$$x - \frac{1}{4}(x^2 - 2) = x \Leftrightarrow x^2 - 2 = 0 \Leftrightarrow x_1 = -\sqrt{2} \text{ or } x_2 = \sqrt{2}.$$

We found 2 fixed points: $\eta_1^* = -\sqrt{2}$ and $\eta_2^* = \sqrt{2}$.

In order to study the stability we compute first

$$f'(x) = 1 - \frac{1}{4} \cdot 2x = 1 - \frac{x}{2}. \quad \text{So}$$

$$|f'(-\sqrt{2})| = \left|1 + \frac{\sqrt{2}}{2}\right| > 1 \quad \text{and} \quad |f'(\sqrt{2})| = \left|1 - \frac{\sqrt{2}}{2}\right| < 1.$$

We conclude that $\eta_1^* = -\sqrt{2}$ is unstable, while

$\eta_2^* = \sqrt{2}$ is an attractor.

b) Denote, as usual, the unique solution of this IVP by $\varphi(k, \eta)$. Note also that the equation is $x_{k+1} = f(x_k)$.

Since $\eta_1^* = -\sqrt{2}$ is a fixed point of f we have

$$\varphi(k, -\sqrt{2}) = -\sqrt{2}, \quad \forall k \geq 0.$$

Analogously, $\varphi(k, \sqrt{2}) = \sqrt{2}, \quad \forall k \geq 0.$

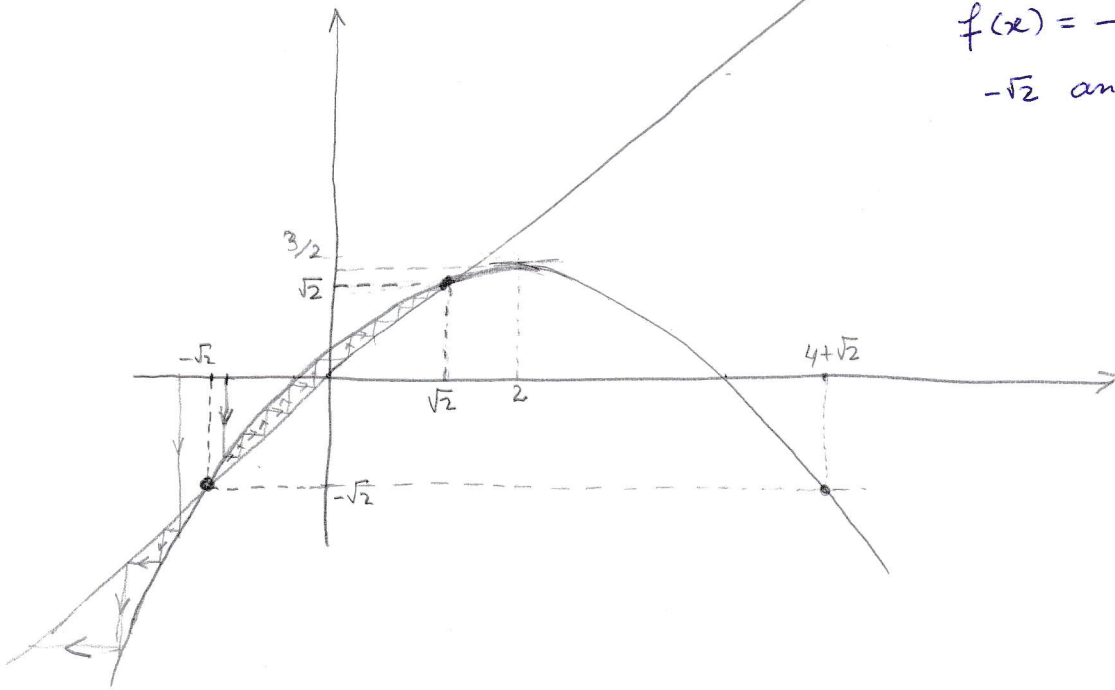
Since $\eta_2^* = \sqrt{2}$ is an attractor for f we have that,

for η sufficiently close to $\sqrt{2}$, $\lim_{k \rightarrow \infty} \varphi(k, \eta) = \sqrt{2}.$

c) We need first to represent the graph of f and $y=x$.

Note that the graph of f is a parabola that intersect the line $y=x$ in the points $(-\sqrt{2}, -\sqrt{2})$ and $(\sqrt{2}, \sqrt{2})$; intersect the line $y=0$ at $(x_1, 0)$ and $(x_2, 0)$ with $x_{1,2} = 2 \pm \sqrt{6}$; and has a maximum at $(2, \frac{3}{2})$. Also, solving the equation

$f(x) = -\sqrt{2}$ we find $-\sqrt{2}$ and $(4+\sqrt{2})$.



We notice that:

- for $x_0 = \eta \in (-\infty, -\sqrt{2})$ we have $\lim_{k \rightarrow \infty} \varphi(k, \eta) = -\infty$ and is strictly decreasing.

- for $x_0 = \eta \in (-\sqrt{2}, \sqrt{2})$ we have $\lim_{k \rightarrow \infty} \varphi(k, \eta) = \sqrt{2}$ and is strictly increasing

- for $x_0 = \eta \in [\sqrt{2}, 2]$ we have $\lim_{k \rightarrow \infty} \varphi(k, \eta) = \sqrt{2}$ and is strictly decreasing

- for $x_0 = \eta \in (2, 4+\sqrt{2})$ we have $x_1 = f(x_0) \in (-\sqrt{2}, 2)$ so the sequence $(\varphi(k, \eta))_{k \geq 1}$ behaves like in the previous cases.

- for $x_0 = 4+\sqrt{2}$ we have $\varphi(k, \eta) = -\sqrt{2}$, $\forall k \geq 1$.

We say that $4+\sqrt{2}$ is eventually the fixed point $-\sqrt{2}$.

Note also that the equation $f(x) = \sqrt{2}$ has ~~2~~ 2 roots, $\sqrt{2}$ and $4-\sqrt{2}$. Hence $4-\sqrt{2}$ is eventually the fixed point $\sqrt{2}$ and $\varphi(k, 4-\sqrt{2}) = \sqrt{2}$, $\forall k \geq 1$.

- for $x_0 = \eta \in (4+\sqrt{2}, \infty)$ we have $x_1 = f(x_0) \in (-\infty, -\sqrt{2})$ so the sequence $(\varphi(k, \eta))_{k \geq 1}$ behaves like in the first case.

We conclude that: $A_{\sqrt{2}} = (-\sqrt{2}, 4+\sqrt{2})$, that is

$$\lim_{k \rightarrow \infty} \varphi(k, \eta) = \sqrt{2}, \text{ for any } \eta \in (-\sqrt{2}, 4+\sqrt{2}). \quad \varphi(k, -\sqrt{2}) = \varphi(k, 4+\sqrt{2}) = -\sqrt{2}, \forall k \geq 1$$

and $\lim_{k \rightarrow \infty} \varphi(k, \eta) = -\infty$, for any $\eta \in (-\infty, -\sqrt{2}) \cup (4+\sqrt{2}, +\infty)$.

Exercise 2 We consider the map $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2 - 1$.

- Find the fixed points of f and study their stability.
- Find the 2-periodic ~~points~~ ^{orbits} of f and study their stability.
- Using the stair-step diagram depict the orbits corresponding to the initial states $\eta = 2$ and, respectively, $\eta = -\frac{1}{4}$ and describe their behaviour.

Solution a) $f(x) = x \Leftrightarrow x^2 - 1 = x \Leftrightarrow x^2 - x - 1 = 0 \Leftrightarrow x_{1,2} = \frac{1 \pm \sqrt{5}}{2}$

We found 2 fixed points: $\eta_1^* = \frac{1-\sqrt{5}}{2}$ and $\eta_2^* = \frac{1+\sqrt{5}}{2}$.

$$f'(x) = 2x$$

$$\left| f'\left(\frac{1-\sqrt{5}}{2}\right) \right| = |1-\sqrt{5}| = \sqrt{5}-1 > 1 \text{ and } \left| f'\left(\frac{1+\sqrt{5}}{2}\right) \right| = 1+\sqrt{5} > 1.$$

Hence both fixed points are unstable.

$$\begin{aligned} \text{b) } f^2(x) &= f(f(x)) = f(x)^2 - 1 = (x^2 - 1)^2 - 1 \Rightarrow \\ &= \cancel{x^4 - 2x^2} = x^4 - 2x^2 \text{ and } (f^2)'(x) = 4x^3 - 4x \end{aligned}$$

$$f^2(x) = x \Leftrightarrow x^4 - 2x^2 - x = 0 \Leftrightarrow$$

We know that the two fixed points of f must be fixed points of f^2 . Thus $(x^2 - x - 1)$ divide exactly the above polynomial.

$$x^2(x^2 - x - 1) + x^3 + x^2 - 2x^2 - x = 0 \Leftrightarrow (x^2 - x - 1)(x^2 + x) = 0$$

$$\Leftrightarrow x_{1,2} = \frac{1 \pm \sqrt{5}}{2}, \quad x_3 = 0, \quad x_4 = -1.$$

So, f^2 has 4 fixed points. Two of them are the fixed points of f and the other two form the 2-periodic orbit: $0, -1, 0, -1, 0, -1, \dots$

We have $(f^2)'(0) = 0$, hence the 2-periodic orbit is an attractor.

c) We left as homework.

Exercise 3 Property Assume that $f \in C^1(\mathbb{R})$ and that $\{\eta_1, \eta_2\}$ is a 2-periodic orbit of f .

If $|f'(\eta_1) \cdot f'(\eta_2)| < 1$ then this orbit is an attractor.

If $|f'(\eta_1) \cdot f'(\eta_2)| > 1$ then this orbit is unstable.

Proof Since $f^2(x) = f(f(x))$ we have

$$(f^2)'(x) = f'(f(x)) \cdot f'(x).$$

Since $\{\eta_1, \eta_2\}$ is a 2-periodic orbit we have

$$f(\eta_2) = \eta_1 \quad \text{and} \quad f(\eta_1) = \eta_2. \quad \text{Anyway we have}$$

$$(f^2)'(\eta_1) = f'(f(\eta_1)) \cdot f'(\eta_1) = f'(\eta_2) \cdot f'(\eta_1).$$

The conclusion follows by Theorem 1 and the definition of attracting periodic point. $\square$.

Exercise 3 We consider the map $T: [0,1] \rightarrow \mathbb{R}$, $T(x) = 1 - |2x - 1|$.

a) Represent the graph of T . Is $\eta_1^* = 0$ a fixed point?

Find all the fixed points of T .

b) Compute the orbits corresponding to the initial states

$$\eta = \frac{3}{4}, \quad \eta = \frac{3}{8}, \quad \eta = \frac{3}{2^n}, \quad n \geq 2.$$

c) How many solutions have the equations:

$$T(x) = 0; \quad T(x) = 1; \quad T(x) = \frac{1}{2}; \quad T^2(x) = 0; \quad T^2(x) = 1;$$

$$T^2(x) = \frac{1}{2}?$$

d) Compute $T^2(x)$.

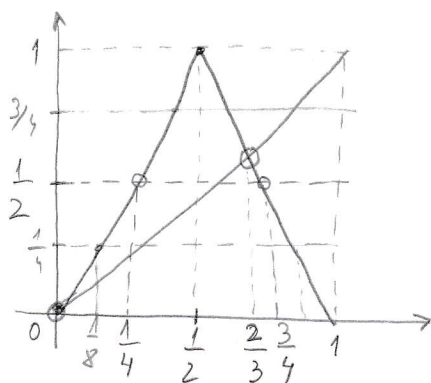
e) Represent the graphs of T^2 and T^3 . How many fixed points they have?

f) T has a 2-periodic orbit? Or a 3-periodic orbit?

T has a 1492-periodic orbit?

Solution

$$T(x) = \begin{cases} 2x, & x \in [0, \frac{1}{2}] \\ 2(1-x), & x \in (\frac{1}{2}, 1] \end{cases}$$



Since $T(0) = 0$, of course $\eta_1^* = 0$ is a fixed point of T .

By solving $2(1-x) = x$ we find that $\eta_2^* = \frac{2}{3}$ is another f.p.

Thus, T has 2 fixed points.

b) $\eta = \frac{3}{4}$, $T(\eta) = T(\frac{3}{4}) = 1 - |2 \cdot \frac{3}{4} - 1| = 1 - \frac{1}{2} = \frac{1}{2}$,

$$T(\frac{1}{2}) = 1, \quad T(1) = 0.$$

So, the orbit is $\frac{3}{4}, \frac{1}{2}, 1, 0, 0, 0, \dots$

$$\frac{3}{8}, \quad T\left(\frac{3}{8}\right) = 1 - \left|2 \cdot \frac{3}{8} - 1\right| = 1 - \left|-\frac{1}{4}\right| = 1 - \frac{1}{4} = \frac{3}{4}$$

So, the orbit is $\frac{3}{8}, \frac{3}{4}, \frac{1}{2}, 1, 0, 0, 0, \dots$

Let $n \geq 3$ then $T\left(\frac{3}{2^n}\right) = 2 \cdot \frac{3}{2^n} = \frac{3}{2^{n-1}}$

$$\frac{3}{2^n} < \frac{1}{2}$$

So, the orbit is $\frac{3}{2^n}, \frac{3}{2^{n-1}}, \dots, \frac{3}{2^3}, \frac{3}{2^2}, \frac{1}{2}, 1, 0, 0, \dots$

c) $T(x) = 0$ has two solutions: 0 and 1

$T(x) = 1$ has one sol.: $\frac{1}{2}$

$T(x) = \frac{1}{2}$ has two sol.: $\frac{1}{4}$ and $\frac{3}{4}$

We deduced all these from the graph.

$$T(T(x)) = 0 \Leftrightarrow T(x) = 0 \text{ or } T(x) = 1 \Leftrightarrow x \in \left\{0, \frac{1}{2}, 1\right\}$$

$$T(T(x)) = 1 \Leftrightarrow T(x) = \frac{1}{2} \Leftrightarrow x \in \left\{\frac{1}{4}, \frac{3}{4}\right\}$$

$$T(T(x)) = \frac{1}{2} \Leftrightarrow T(x) = \frac{1}{4} \text{ or } T(x) = \frac{3}{4} \Leftrightarrow x \in \left\{\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right\}$$

$$d) \quad T^2(x) = T(T(x)) = \begin{cases} 2T(x), & T(x) \in [0, \frac{1}{2}] \\ 2 - 2T(x), & T(x) \in (\frac{1}{2}, 1] \end{cases} =$$

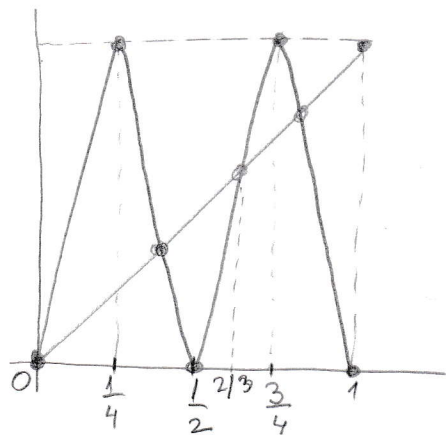
$$= \begin{cases} 4x, & x \in [0, \frac{1}{4}] \\ 2 - 4x, & x \in (\frac{1}{4}, \frac{1}{2}] \\ 2 - 4(1-x), & x \in (\frac{1}{2}, \frac{3}{4}] \\ 4(1-x), & x \in (\frac{3}{4}, 1] \end{cases}$$

e) From c) we know that $T^2(0) = T^2(\frac{1}{2}) = T^2(1) = 0$

and $T^2(\frac{1}{4}) = T^2(\frac{3}{4}) = 1$.

From d) we know that T is linear on each of the intervals $(0, \frac{1}{4})$, $(\frac{1}{4}, \frac{1}{2})$, $(\frac{1}{2}, \frac{3}{4})$ and $(\frac{3}{4}, 1)$.

Hence:



We notice that T^2 has 4 fixed points.

Two of them are, of course, the fixed points of T , 0 and $\frac{2}{3}$.

In order to represent the graph of T^3 we solve the eq.

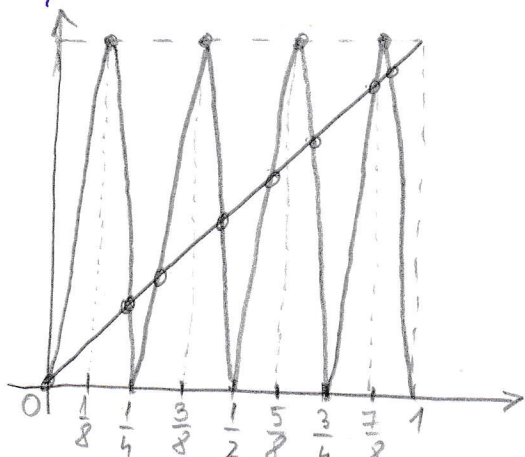
$$T^3(x) = 0 \Leftrightarrow T^2(T(x)) = 0 \Leftrightarrow T(x) \in \{0, \frac{1}{2}, 1\} \Leftrightarrow$$

$$\Leftrightarrow x \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$$

and also the eq

$$T^3(x) = 1 \Leftrightarrow T(T^2(x)) = \frac{1}{2} \Leftrightarrow T^2(x) = \frac{1}{2} \Leftrightarrow x \in \{\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}\}$$

We realize that T^3 is linear on each of the intervals $(0, \frac{1}{8})$, $(\frac{1}{8}, \frac{1}{4})$, $(\frac{1}{4}, \frac{3}{8})$, $(\frac{3}{8}, \frac{1}{2})$, $(\frac{1}{2}, \frac{5}{8})$, $(\frac{5}{8}, \frac{3}{4})$, $(\frac{3}{4}, \frac{7}{8})$, $(\frac{7}{8}, 1)$. Hence



We notice that T^3 has 8 fixed points.

Two of them are, of course, the fixed points of T ,

0 and $\frac{2}{3}$.

$\neq$) Since T^2 has two fixed points which are not fixed points of T , we deduce that they form a 2-periodic orbit for T . Actually, they can be found exactly by solving $2-4x=x \Leftrightarrow x=\frac{2}{5}$ and computing $4-4x=x \Leftrightarrow x=\frac{4}{5}$.

So, T has a unique 2-periodic orbit: $\left\{ \frac{2}{5}, \frac{4}{5} \right\}$.

Since T^3 has 6 fixed points which are not fixed points of T , we deduce that they form two 3-periodic orbits.

Since T has at least a 3-periodic orbit, we deduce using the Sharkowski Theorem that T has periodic orbits of any period.

Exercise 4 We consider the map $G: [0,1] \rightarrow \mathbb{R}$, $G(x) = 4x(1-x)$.

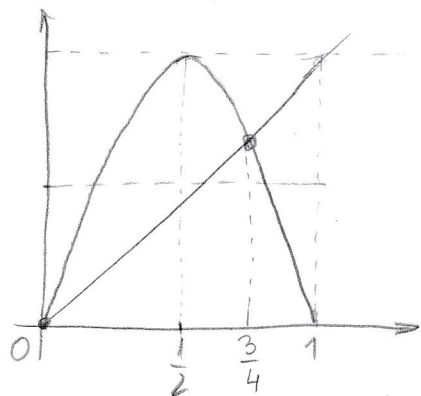
a) Represent the graph of G . Is $x_0^* = 0$ a fixed point? Find all the fixed points of G .

b) How many solutions have the equations:
 $G(x) = 0$; $G(x) = 1$; $G(x) = \frac{1}{2}$; $G^2(x) = 0$, $G^2(x) = 1$; $G^2(x) = \frac{1}{2}$;
 $G^3(x) = 0$; $G^3(x) = 1$?

c) Represent the graphs of G^2 and G^3 . How many fixed points they have?

d) G has a 2-periodic orbit? Or a 3-periodic ~~orbit~~?
 G has a 1492-periodic orbit?

Solution We have $G(0) = G(1) = 0$ and the graph of G is a parabola with a maximum at $(\frac{1}{2}, 1)$.



Since $G(0) = 0$, of course $\eta_1^* = 0$ is a fixed point of G .

We note that G has another f.p. which can be found by solving

$$4x(1-x) = x, \quad x \neq 0$$

$$4 - 4x = 1 \quad x = \frac{3}{4} \quad \eta_2^* = \frac{3}{4}$$

Thus, G has two fixed points 0 and $\frac{3}{4}$.

b) From the graph we see that

$G(x) = 0$ has two solutions: 0 and 1

$G(x) = 1$ has one solution: $\frac{1}{2}$

$G(x) = \frac{1}{2}$ has two solutions: $a_1 \in (0, \frac{1}{2})$ and $a_2 \in (\frac{1}{2}, 1)$

$$G^2(x) = 0 \Leftrightarrow G(G(x)) = 0 \Leftrightarrow G(x) \in \{0, 1\} \Leftrightarrow x \in \{0, \frac{1}{2}, 1\}$$

$$G^2(x) = 1 \Leftrightarrow G(G(x)) = 1 \Leftrightarrow G(x) = \frac{1}{2} \Leftrightarrow x \in \{a_1, a_2\}$$

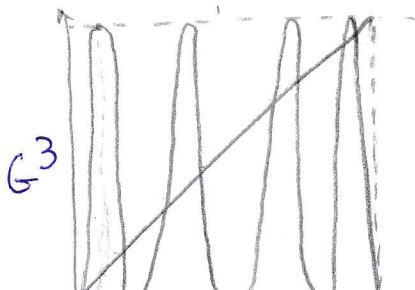
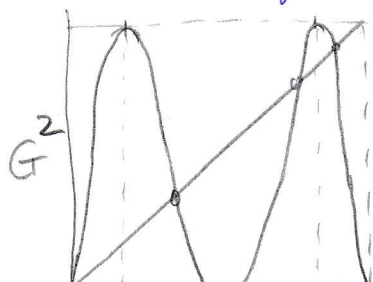
$$G^2(x) = \frac{1}{2} \Leftrightarrow G(G(x)) = \frac{1}{2} \Leftrightarrow G(x) \in \{a_1, a_2\} \quad \text{and this}$$

equation has 4 solutions.

$$G^3(x) = 0 \Leftrightarrow G(G^2(x)) = 0 \Leftrightarrow G^2(x) \in \{0, 1\} \Leftrightarrow x \in \{0, a_1, \frac{1}{2}, a_2, 1\}$$

$$G^3(x) = 1 \Leftrightarrow G(G^2(x)) = 1 \Leftrightarrow G^2(x) = \frac{1}{2} \quad \text{has 4 solutions.}$$

c) The graph of G^2 looks like the graph of T^2 , only that the graph of G^2 is smooth.



The discussions are like for T^2 and T^3 , respectively.